OEM Equipment Manual

Process Cooling & Filtration Systems — Troubleshooting Extract

Manual No.	OEM-MAN-CW3	**Revision**	02
Equipment	CW-3 / FIL-07 / HVAC	**Issued**	01-Jun-2022
Manufacturer	ThermaCore Systems	**Classification**	Reference

Section 4.2 — Chiller CW-3: Reduced Cooling Capacity

Symptom: discharge temperature above 8 °C with the compressor running at full demand.

Probable causes (by frequency):

Condenser fouling; low refrigerant charge; failed expansion valve; fouled plate heat exchanger on the process side.

Recommended checks:

- Verify condenser inlet air is unobstructed and inlet temperature is below 35 °C.
- Read refrigerant suction and discharge pressures against the nameplate table.
- If the plate heat-exchanger differential pressure exceeds 35 kPa, clean-in-place is required before further diagnosis.

Caution: Maximum permitted coolant setpoint reduction rate is 5 percent per 90-second interval. Exceeding this risks thermal shock to the plate pack and voids the warranty on the exchanger assembly.

Section 5.1 — Filter Skid FIL-07: Differential Pressure

Rising dP: indicates progressive blockage. Compare against the bubble-point value recorded at installation. A filter failing bubble point must be replaced before the batch continues.
Falling dP: may indicate a seal breach or bypass. Verify the seal integrity index and inspect for leakage. A sudden dP drop is a containment concern, not merely a process excursion, and requires immediate escalation.

Condition	Indicated Cause	OEM Action
dP > 114 kPa	Filter blockage	Integrity check; replace if bubble-point fails
dP < 96 kPa	Seal breach / bypass	Escalate; inspect seal integrity index
dP oscillating	Exhaust restriction	Check gas exhaust flow and condensate trap

Section 6.3 — HVAC Air Handling: Humidity Control

The cooling coil is the primary means of dehumidification. Coil approach temperature elevated by more than 3 °C indicates fouling or reduced chilled-water flow and will prevent the unit from holding the RH band.
HEPA differential pressure above 92 percent of the loading limit restricts airflow and manifests as an RH excursion. Replace HEPA banks at the loading limit, not on a fixed calendar interval.

Component	Normal	Service Trigger
Cooling coil approach	< 2 °C	> 3 °C — clean coil
HEPA dP	< 250 Pa	> 92% loading — replace
Supply fan	design flow	> 6% deviation — investigate